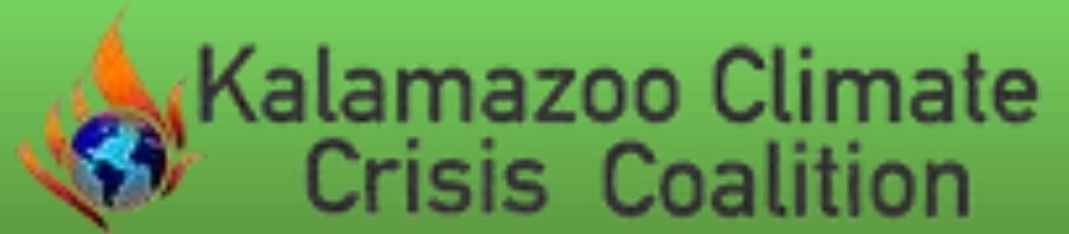




Ardea Climate 101

- Report on Climate August 2026
- Understanding the Climate Emergency
- Taking Action

Dr. Allen Webb
allen.webb@wmich.edu



My Climate Action



Activism

Teaching



Writing & Speaking





Western Michigan University

Professor of English Education
and Postcolonial Studies

Dr. Allen Webb

Courses

Resources

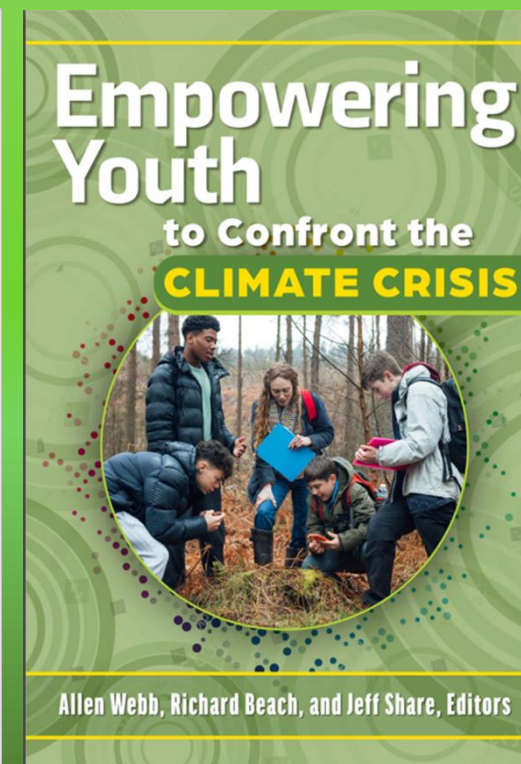
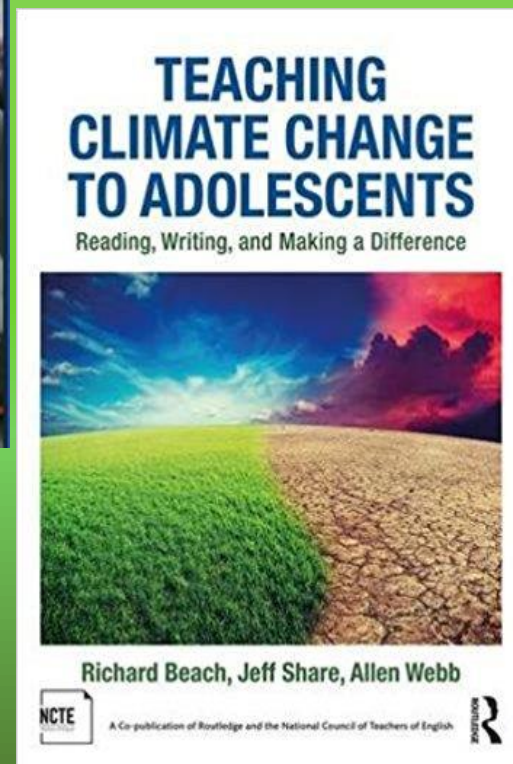
Research

Contact

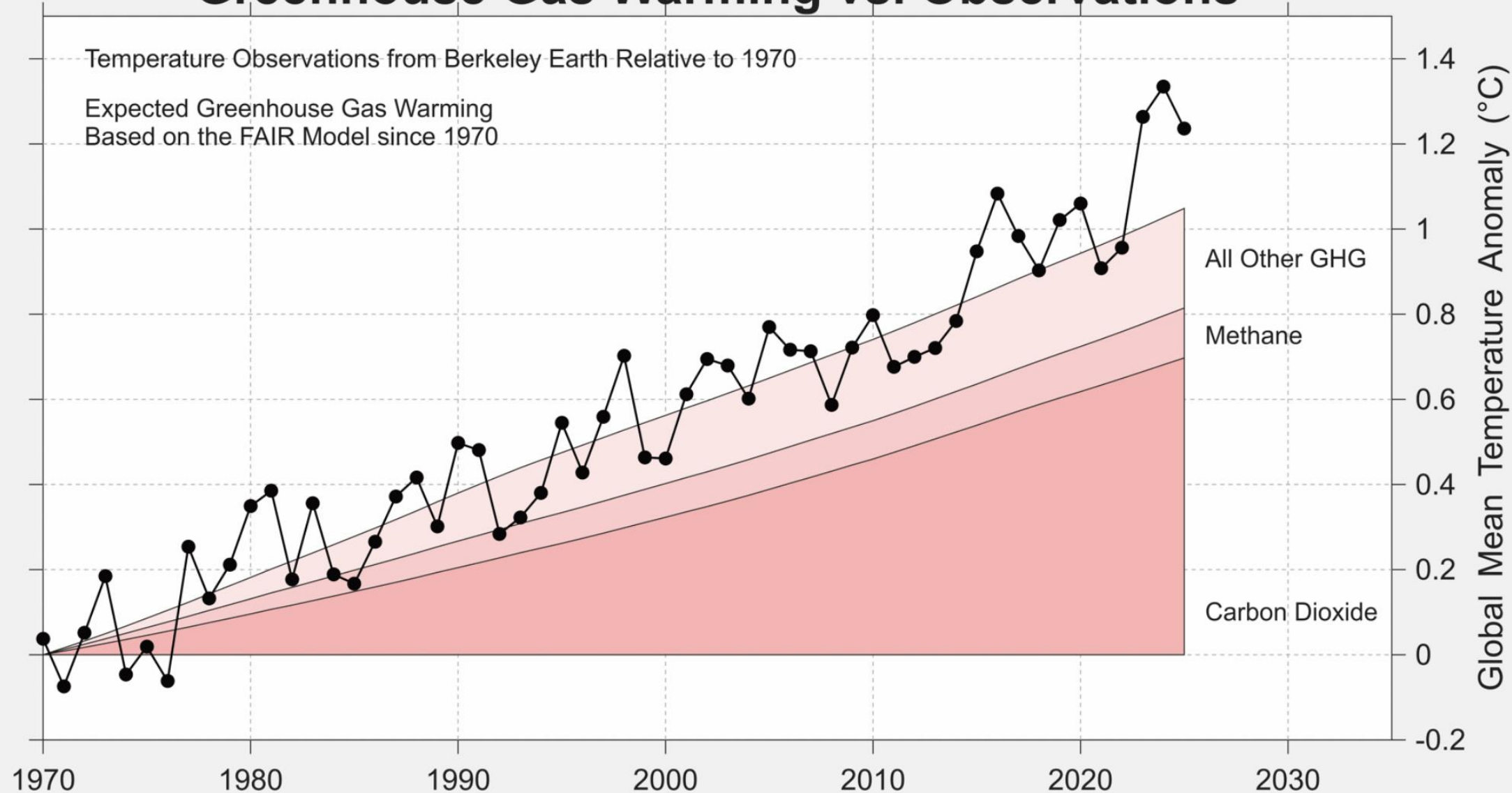
My scholarship and teaching focus on English education, postcolonial and minority literature, Internet learning, and climate change from a humanities perspective.

My site contains on-line course syllabi, resources for preservice and inservice English teachers, information about my research and publications, and contact and personal data.

AllenWebb.net



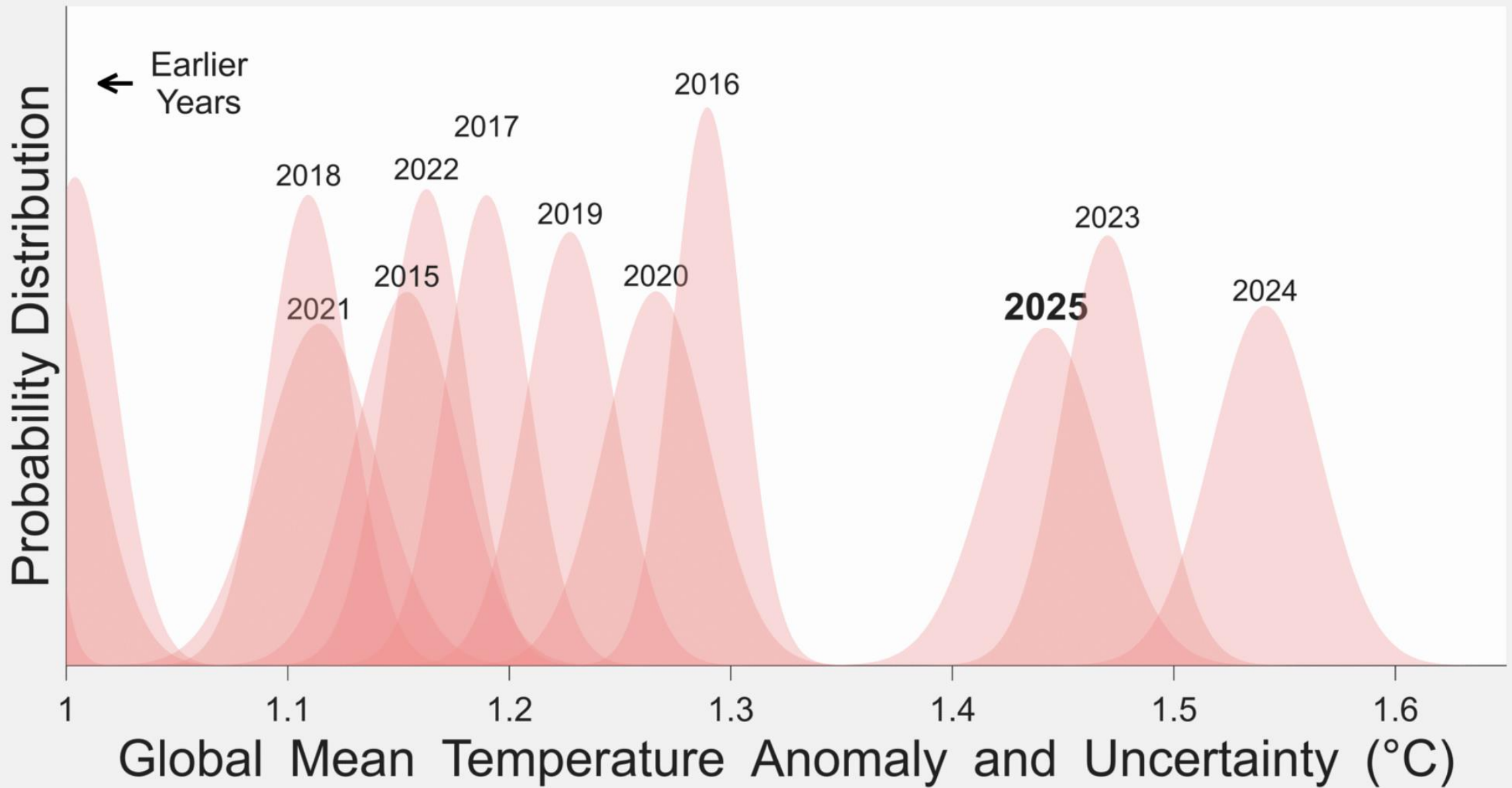
Greenhouse Gas Warming vs. Observations



2023-26 Hottest Years in 120,000 years...

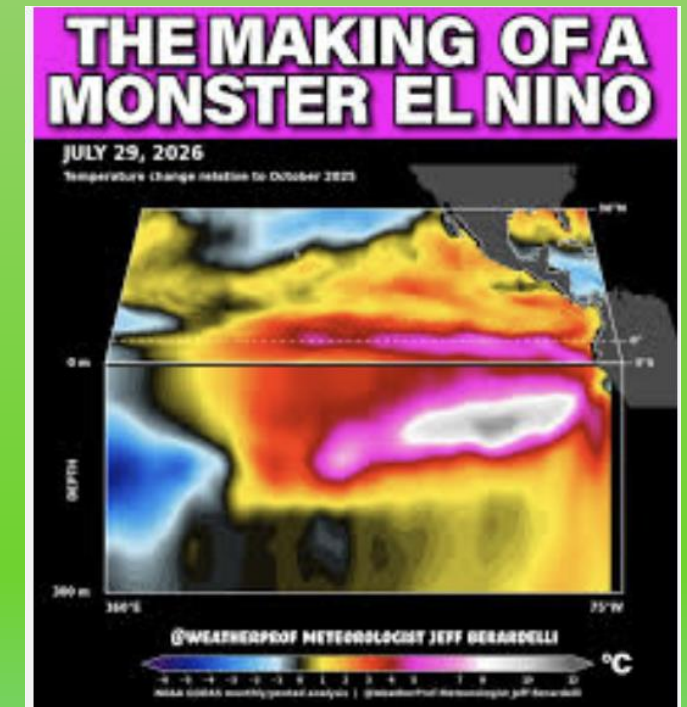
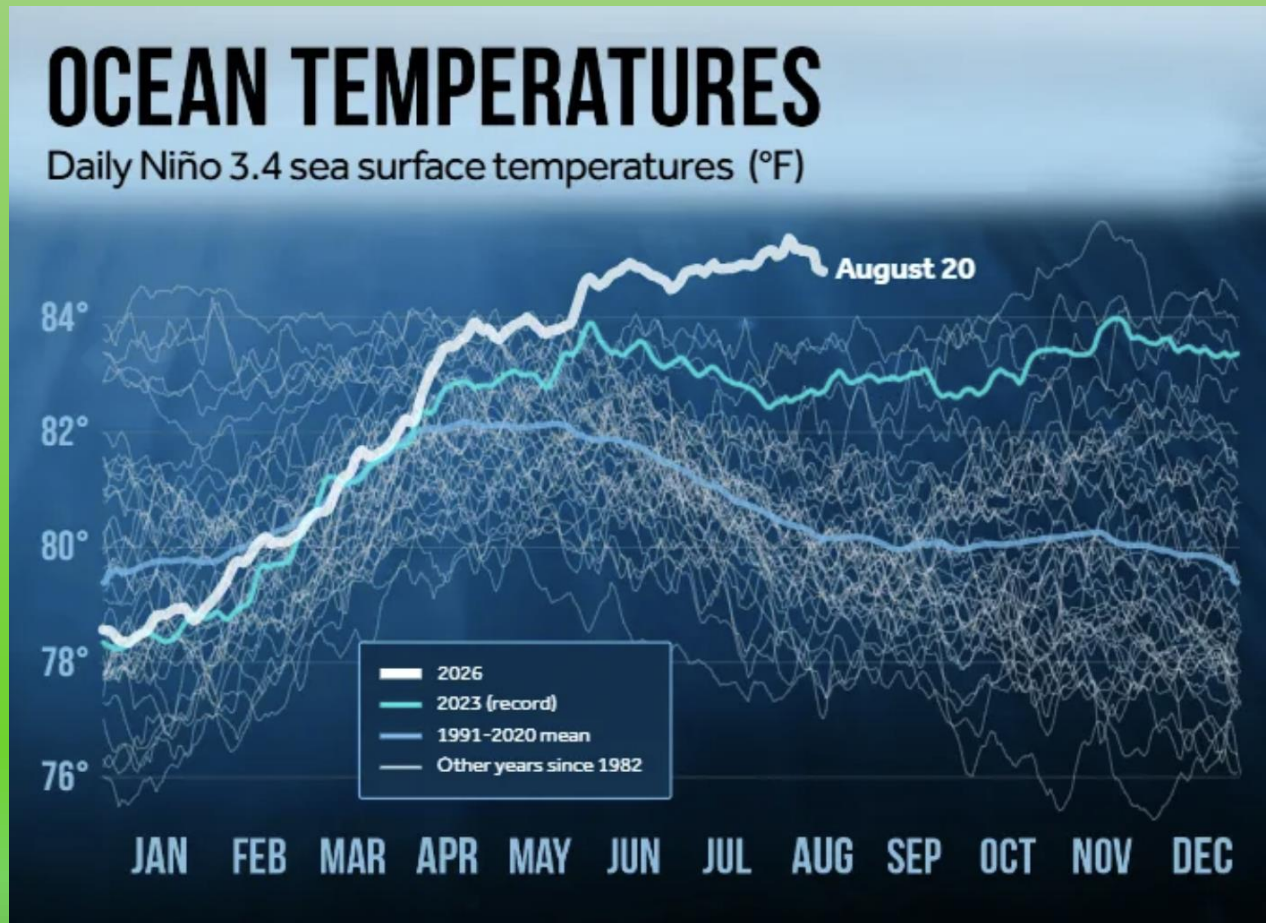


- 2025 third warmest year ever recorded after 2023 & 2024.
- 2023-5 jump above already rapidly increasing warming.
- 2023 & 2024 exceed the 2015 Paris Agreement target of 1.5°C. Several months of 2024 >2°C. 2025, “La Niña year,” 1.44°C
- Global warming is accelerating: “Past warming rate is no longer a reliable predictor of the future” (Berkeley Earth)



Based on Berkeley Earth's estimates of the global annual average temperature increase relative to 1850-1900. Each year's individual uncertainty is shown, but does not include the systematic uncertainty in the baseline.

2026-2027 Monster El Niño - Already Arrived



Rapidly increasing in strength

May raise global temperatures .3°C to 1.75°C in 2027?

2025 Global Deaths Due to Climate Change

Extreme heat	~300,000–400,000
Wildfire smoke	~75,000–150,000
Floods, storms & extreme precipitation	~20,000–50,000
Drought, crop failure & food insecurity	~50,000–150,000
Climate-sensitive infectious disease	~25,000–75,000
Other climate-related effects	~25,000–75,000

$700,000/120=5833$

Total: 500,000– 1,000,000

Climate Emergency: 1) Greenhouse gas emissions!



Burning Fossil Fuels



Deforestation



Animal Agriculture

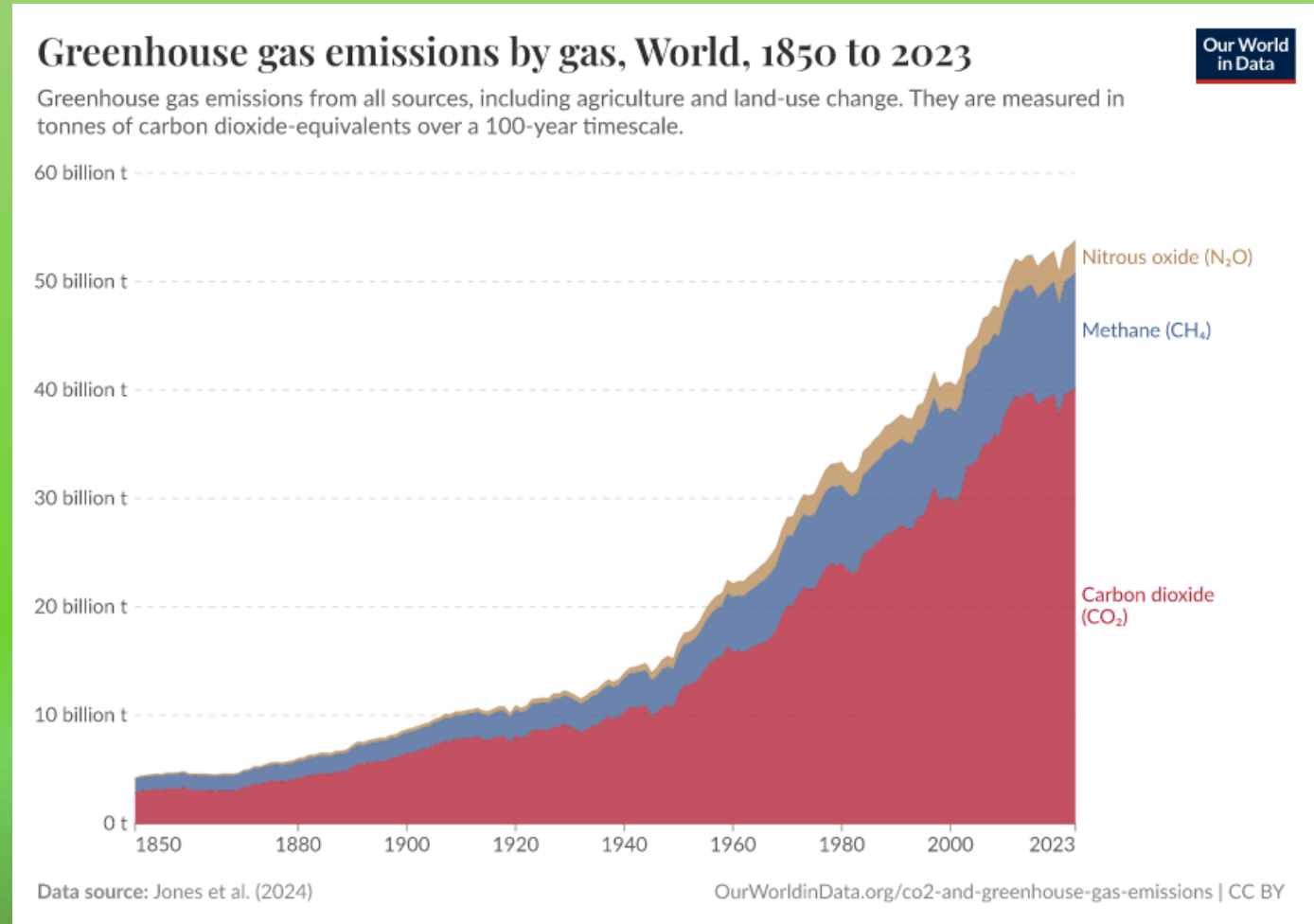
2) Failure of governments to act!

United Nations Framework
Convention on Climate Change
(UNFCCC) 1992

– 33 Years Ago –

Since 1992 we've emitted as much
CO₂ as in the entire industrial era
back to the 1700's.

Greenhouse Gas Emissions keep
rising! [2023](#), [2024](#), 2025, 2026
setting or matching record highs.



The Climate is Breaking Down



- High temperatures
- Droughts
- Wildfires
- Storms
- Flooding
- Polar Vortex
- Ocean Currents
- Species Loss
- Refugees

What we are seeing is
NOT the “New Normal”



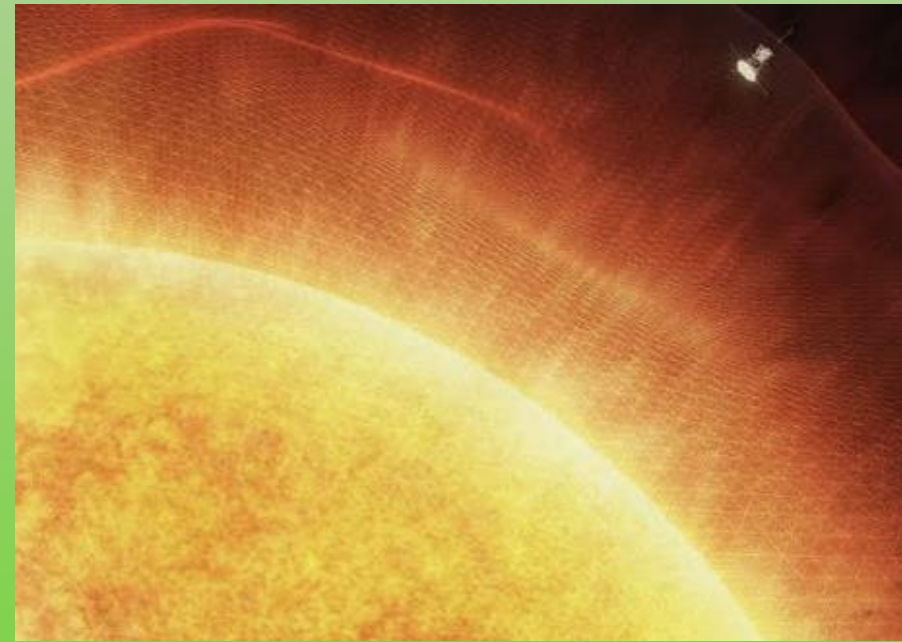
What's coming is far more serious than what we see now!

Climate change does not just impact “nature.” It impacts us!

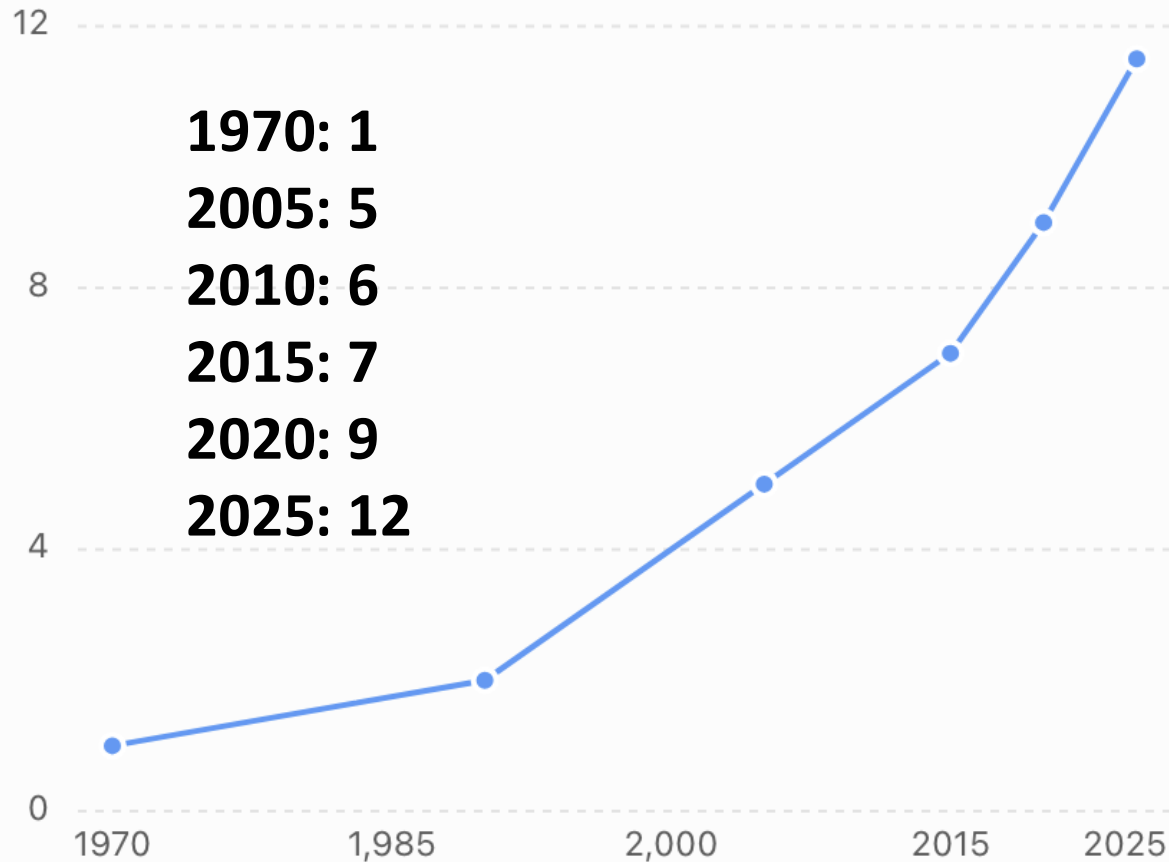
- Our cities, our food, our homes & buildings, the breathability of our air
- Livable temperatures: above wet-bulb 95°F (35°C) limit for human survival
- Mass migration of billions of people
- Stability of society, collapsing laws, governments, borders, conflicts

The 1945 Hiroshima Bomb

- Equal to 15,000 tons of conventional explosives
- Released 63 trillion joules of energy
- Devastated 5 square miles
- Far hotter than the surface of the sun - millions of °C
- 70-80,000 people died instantly, 60,000 afterward



Rate human-released greenhouse gasses are warming Earth in Hiroshima Bombs per Second (HpS)



1,036,800

Testifying

- DC student trip
- Local testifying

Humans produce a lot of



- In 2024 humans released [41.6 billion](#) metric tons of CO₂, 4.5 times the volume of Lake Michigan.
- The cumulative weight of all CO₂ emitted by humanity significantly outweighs the total mass of all physical things currently standing on earth
- Human CO₂ waste produced since the Industrial Revolution (2.5 trillion metric tons) weighs **more than double** the physical infrastructure humanity (1.15 trillion metric tons) has ever built.

Driving a gas-powered car produces a lot of



- The weight of CO₂ produced by burning gasoline is 3.1 times greater than the weight of the gasoline alone because each carbon atom combines with 2 oxygen atoms (each 33% heavier than the carbon atom), and gasoline is 85% carbon by weight.
- Since one gallon of gasoline weighs 6 pounds, burning the gasoline of a 14-gallon tank (car) releases **260 pounds** of CO₂ ($6 \times 14 \times 3.1$), and a 30-gallon tank (pick up truck) releases **558 pounds** of CO₂.
- At 72°F 1 pound of CO₂ is 8.8 ft³. For a car that would be 2288 ft³ or truck 4910 ft³, or about 4,600 or 10,000 party balloons.

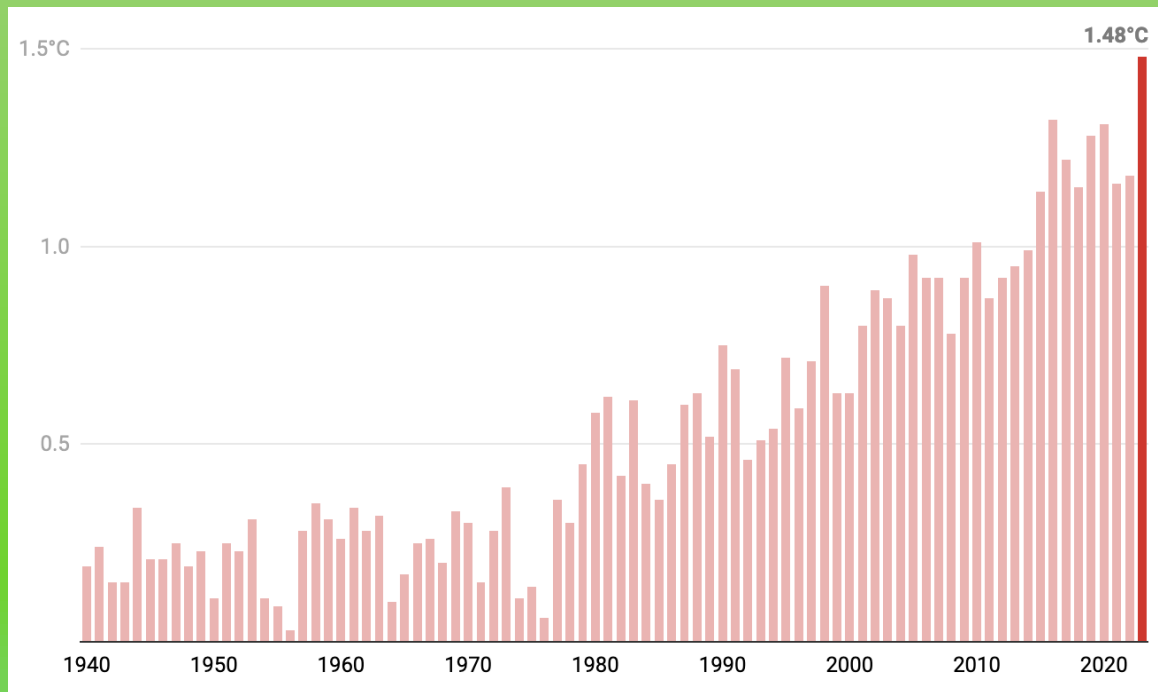


Fossil fuels responsible for 1 in 5 deaths!

- Air pollution caused by the burning of fossil fuels responsible for 8.7 *million premature deaths globally in 2018*
- Especially in areas with high fossil fuel use, North America, Europe, China, & India



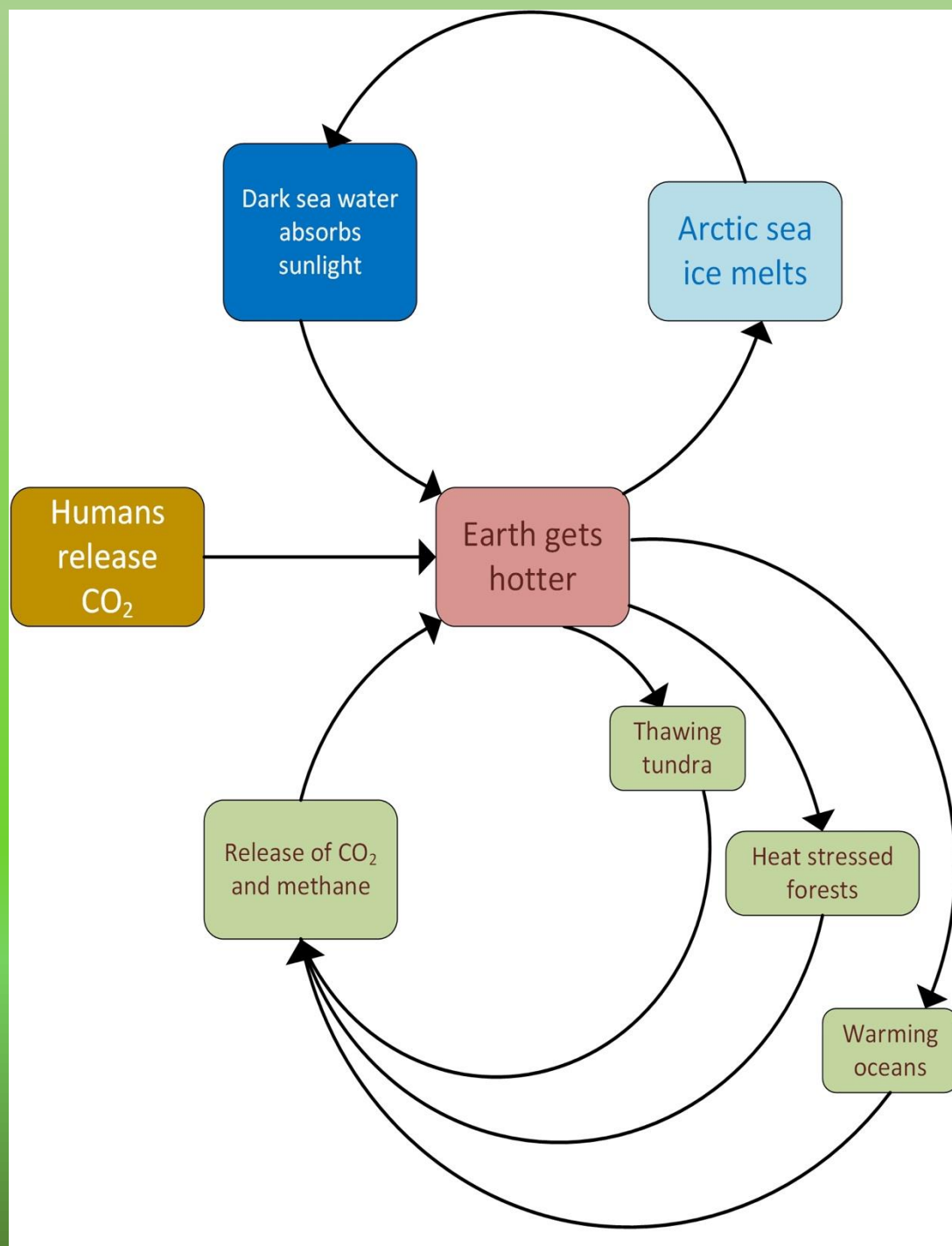
Exponential Growth Can Confuse Us



1st month 1 lily pad. 2 months 2 pads. Covered in 36 months. [When is the pond half covered?](#)

We are entering a “critical and unpredictable new phase of the climate crisis”

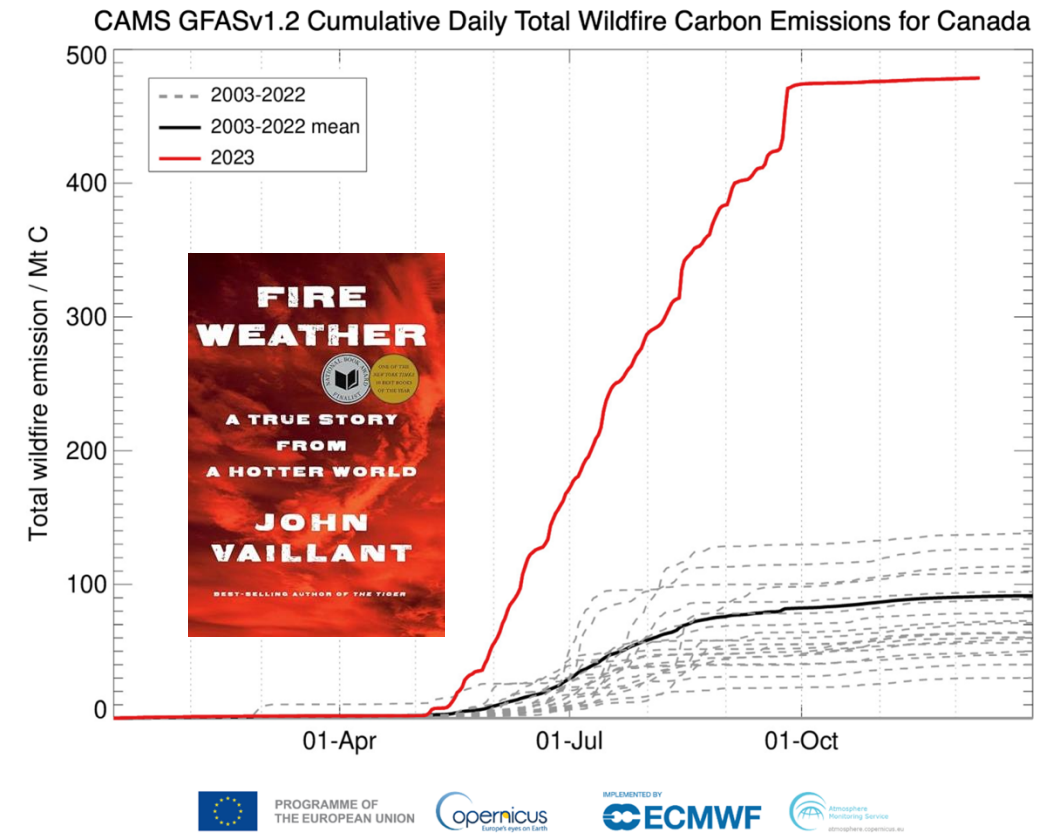
- Trees and land -- earth's carbon sinks- no longer absorbing carbon.
- World rivers drying up. Amazon “savannafication.”
- Thwaites “doomsday glacier” undermined by warmer water.
- 28 positive feedback loops, including increasing emissions from melting permafrost, can trigger multiple tipping points, inc AMOC currents.
- “Climate change has already displaced millions of people, with the potential to displace hundreds of millions or even billions. That would likely lead to greater geopolitical instability, possibly even partial societal collapse.”



2°C, even 1.5°C, may initiate dangerous feedback loops & tipping points that could spiral beyond human control...

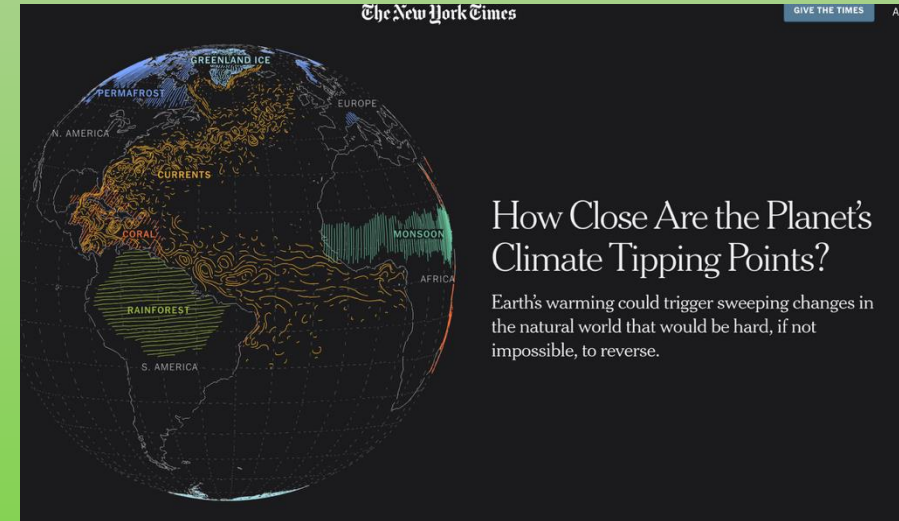
Wildfires turn Canada's vast forests from carbon sink into super-emitter

- In 2023 about 2.2 billion tons of CO2 released, also fostering melting of permafrost
- Same as 4 x global airline industry in a year, or 647 million gas burning cars
- Canadian wildfires only 22% of 2023 global total wildfires



1. Coral Reefs
2. Permafrost
3. Greenland Ice Sheet
4. West Antarctic Ice
5. West African Monsoon
6. Amazon Rainforest
7. Atlantic Currents

NY Times
Aug 2024



<https://www.nytimes.com/interactive/2024/08/11/climate/earth-warming-climate-tipping-points.html>

No Longer Any “Carbon Budget”

- Global heating has already passed the danger point.
- Fossil fuels must stay in the ground; exploration, production, and burning must stop.
- Fossil fuel states and companies are resisting removing trillions of dollars of assets from their balance sheets and imagined future profits.
- “It is hard for a man to understand something if his salary is based on not understanding.” - Upton Sinclair
- “By 1.5°C there may be so many carbon-cycle and other feedbacks under way and active systems instabilities that the climate will not stabilize at 1.5°C. “ - David Spratt, National Center for Climate Restoration, August 2025



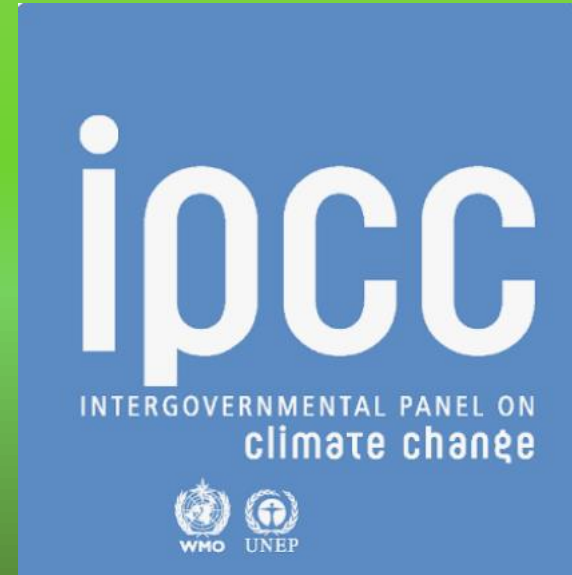
IPCC Standards Failing and Inadequate

(Intergovernmental Panel on Climate Change)

To keep the world livable and avoid risks of feedback loops there must be a reduction of 45% of GHG by 2030, and “net-zero”* by 2050

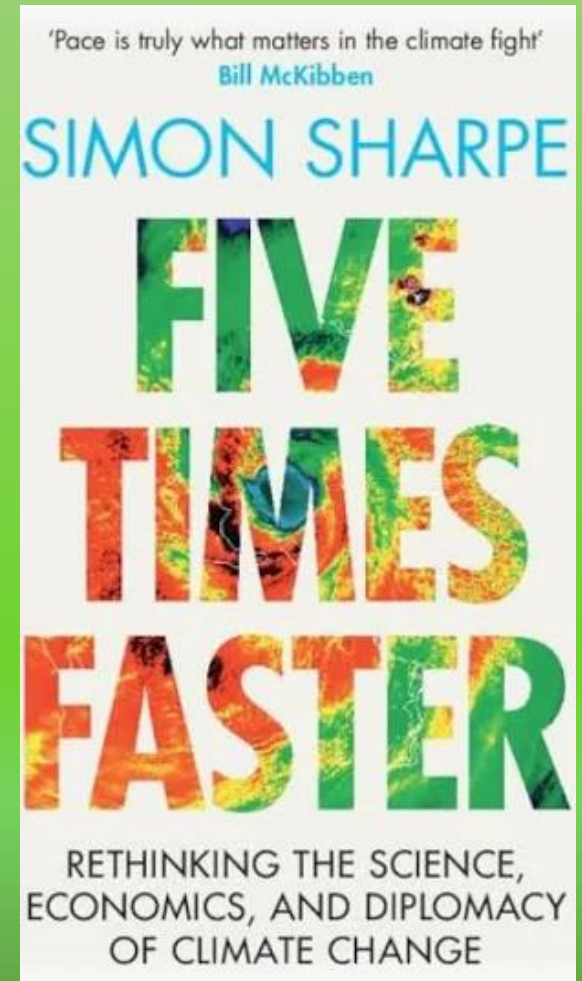
Current Paris Agreement (voluntary, self-reported) nationally determined contributions (NDCs) are still for 9% increase by 2030.

*Assumes substantial Carbon Capture and Storage (CCS) which is impossible.



Prediction vs Risk Assessment

- Prediction focuses on providing at least 90% accurate estimates of future events
- Risk Assessment focuses on worst case scenarios and their likelihood
- Risks of climate change keep increasing until all greenhouse gas emissions are stopped and the world returns to equilibrium
- Medium level of GHG emissions = 20% chance of 4°C
- Would you take an airplane flight with a 20% chance of crashing?



Future Impacts of Increasing Temperature

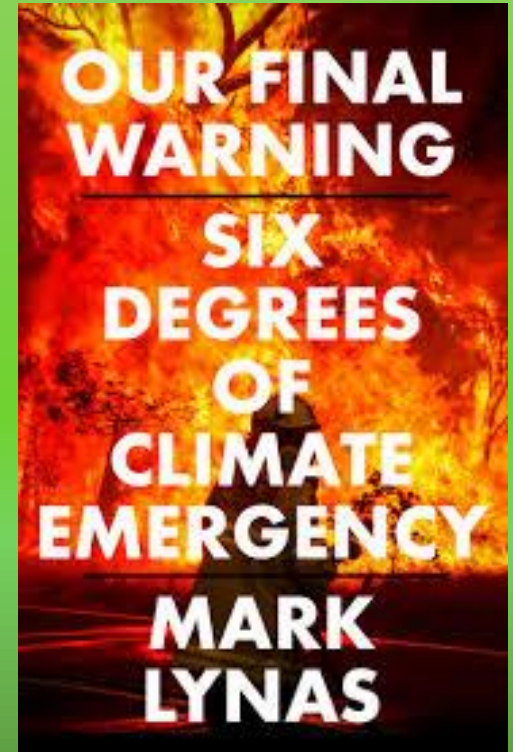
- 3°
 - Tremendous heat waves. 1/3 of global land area exceeds heat deadly threshold 20 days per year
 - Mediterranean region becomes desert, drought engulfs majority of land area now inhabited by humans, megafires
- 4°
 - Collapse of food production in China, the western half of India, southern Africa, Australia, and the western United States
 - Sea level rise forces billions to move

“The difference between 2°C and 4°C is human civilization.”

[Hans Schellnhuber](#), founding director
of the Potsdam Institute for Climate Impact Research

“If we stay on the current business-as-usual trajectory, we could see two degrees as soon as the early 2030s, three degrees around mid-century, and four degrees by 2075 or so. If we’re unlucky with positive feedbacks...from thawing permafrost in the Arctic or collapsing tropical rainforests, then we could be in for five or even six degrees by century’s end.”

Every tenth of a degree matters — and warming in the pipeline means delay = disaster!



Earth's 6 Mass Extinction Events

450 M 75%*	global heating: greenhouse gas
380 M 75%	global heating: greenhouse gas
252 M 96%	global heating: greenhouse gas
200 M 80%	global heating: greenhouse gas
66 M 75%	asteroid & greenhouse gas
Present	global heating: greenhouse gas



- Currently adding carbon at a rate 10x faster than at any time in Earth's history.
- Already more carbon in the atmosphere than the last 20 million years

*percentage extinction of all life on earth



The “Oh, Shit” Moment

They say that everyone who finally gets it about climate change has an “Oh, shit” moment — an instant when the full scientific implications become clear and they suddenly realize what a horrifically dangerous situation humanity has created for itself. - Mark Hertsgaard, [*The Nation*](#)

Al Gore once told me that he thinks that everyone who cares deeply about *climate change* has had what he called an "*oh, shit*" *moment* when the full scientific implications become clear and the future scary. - Jeff Goodell

What can be done?

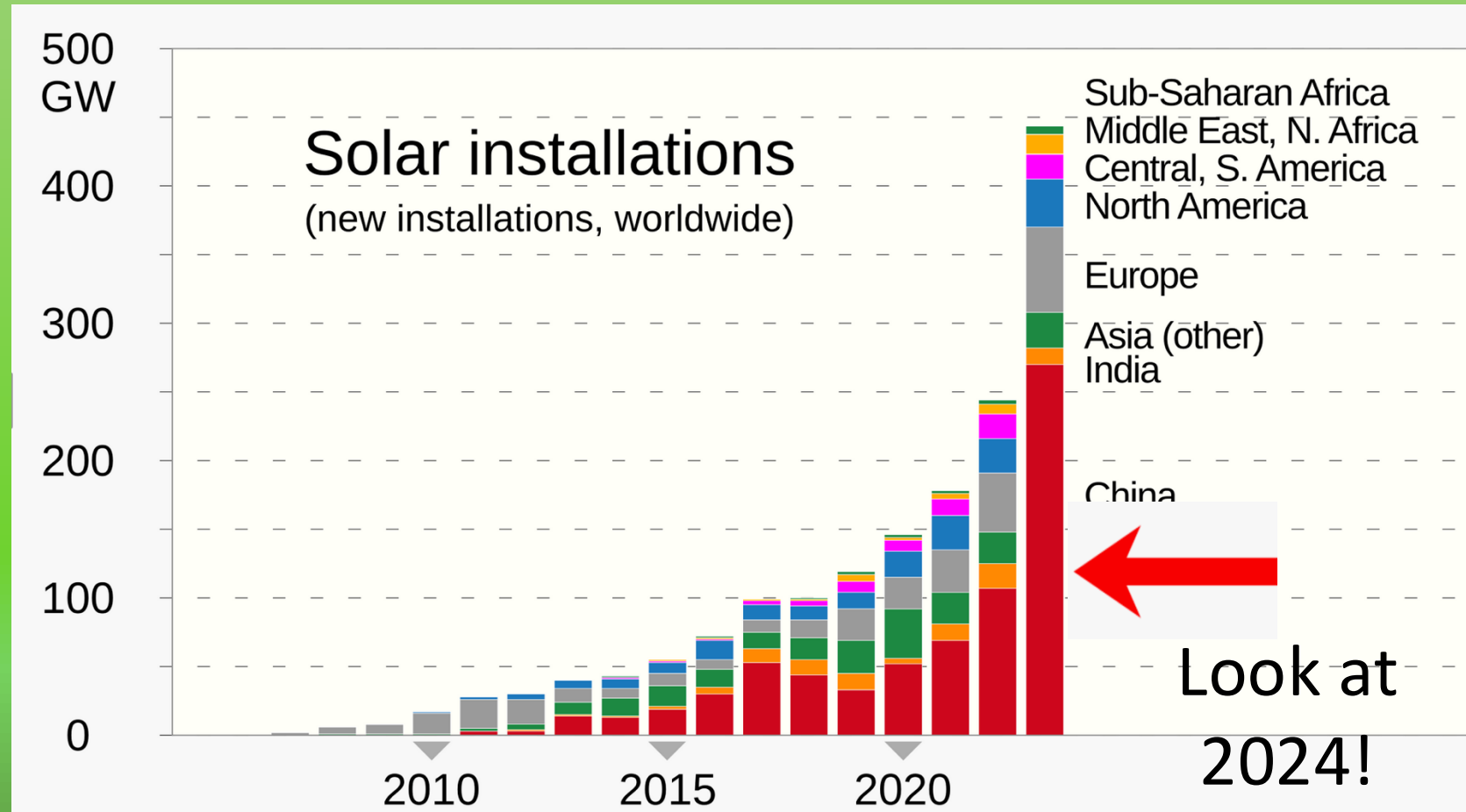
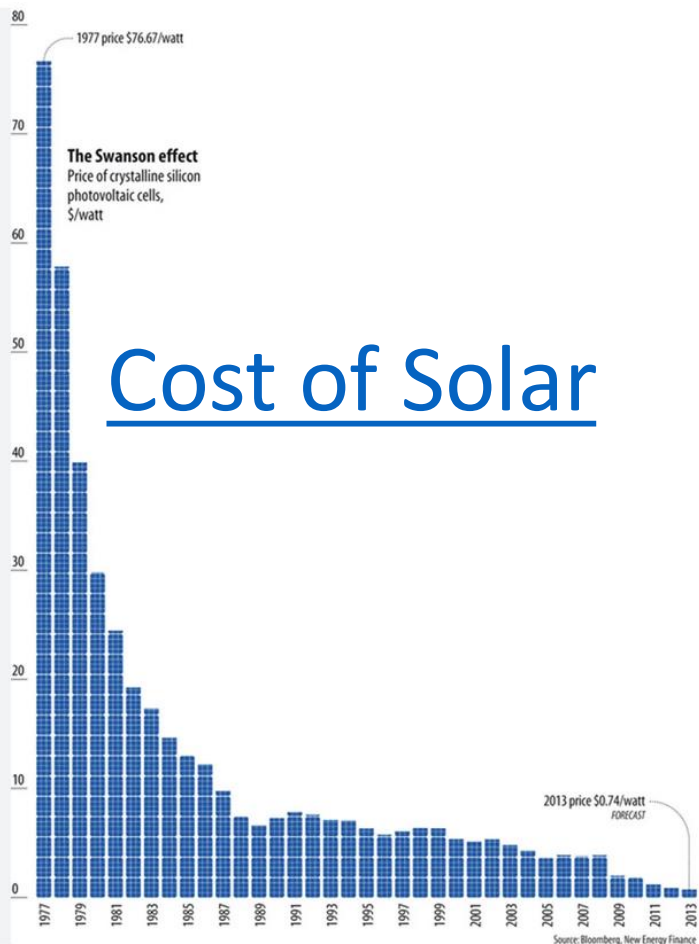
Transformation Not Efficiency



- Environmental thinking from the 1970s must shift to 21st C realities.
- Individuals conserving, reducing, recycling is not going to get us there.
- The system must be transformed to **end all greenhouse gas emission**.
- Reducing meat consumption, especially beef, insulating homes and buildings, less air travel, less consumption – fine and good, but energy, agriculture, transportation, and industrial systems must be changed.

Two Charts Show Transformation Beginning

Cost of Solar



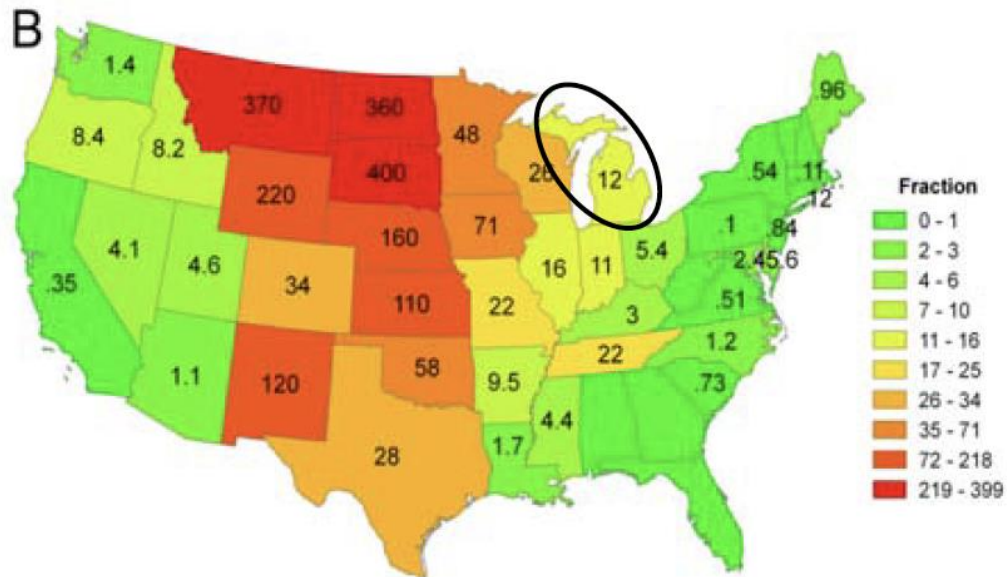
Renewable Energy Breakthrough

- [Solar](#) power is being deployed 100x faster than nuclear.
- In [California solar and wind](#) supplied 100% of power, for months.
- China [6 years](#) ahead of schedule to build renewables.
- Far greater efficiency and less cost than any other energy sources.

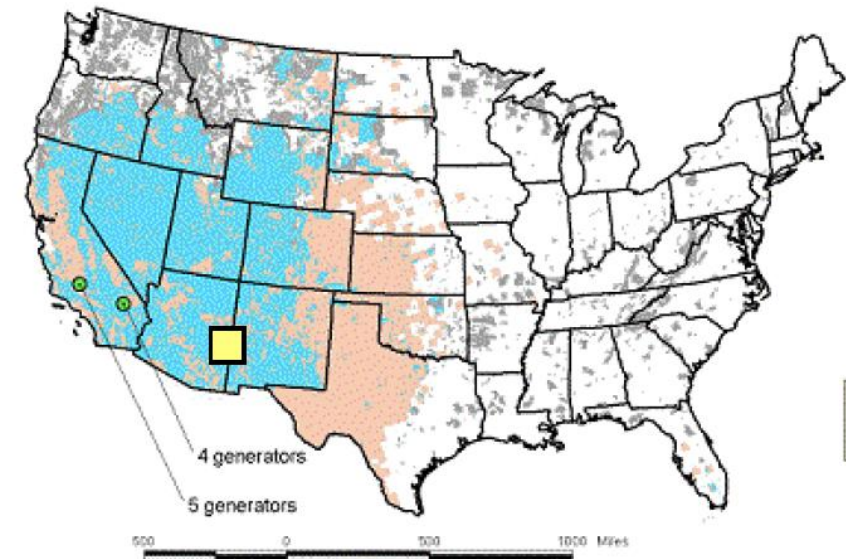


Wind, solar, and batteries can provide **all** US power!

Wind power could supply 16 times current U.S. electricity demand using only onshore windmills



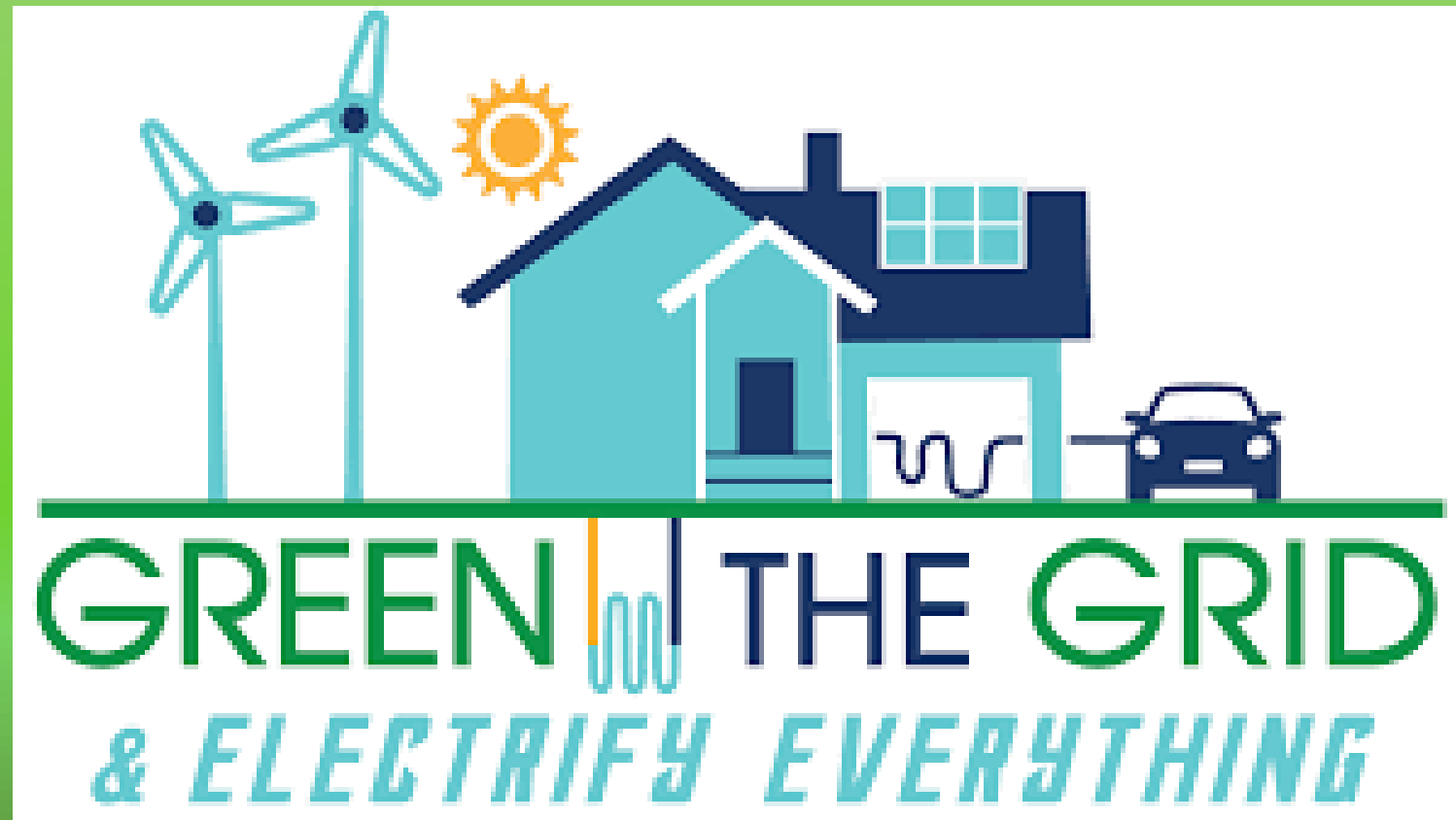
A CSP solar array 100 x 100 miles could provide all of U.S. electricity needs today



- excess energy captured during the day could be stored as heat and used to produce electricity at night

[Mark Jacobson, Stanford](#)

Electrify. Everything.



"Required reading for an economy-wide
green transition in the USA."
MARIANA HAZIVICATO, AUTHOR OF MISSION ECONOMY

ELECTRIFY

AN
OPTIMIST'S PLAYBOOK
FOR OUR
CLEAN ENERGY
FUTURE

SAUL GRIFFITH

Problematic

Carbon tax

NOT Solutions

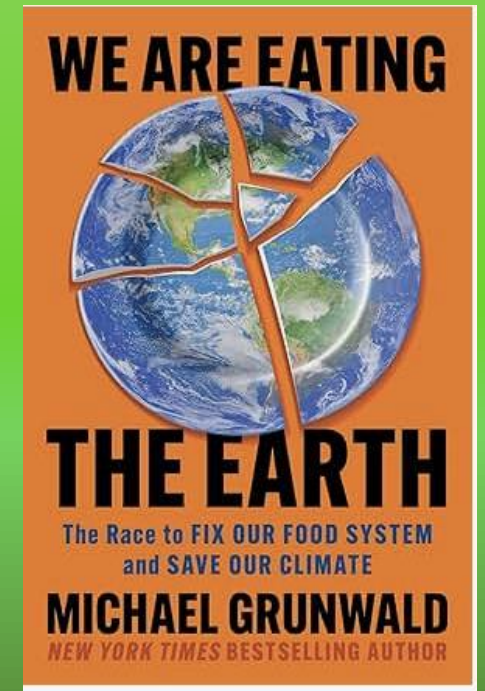
- Natural (fossil) gas
- Fracking
- Carbon capture & storage
- Hydrogen, inc “green”
- Nuclear
- Geoengineering
- Ethanol
- Biomass combustion



Someone profits from these “not solutions”... but not the rest of us or life on earth.

Ethanol – 25% More Greenhouse Gases than Gasoline!

- In the US gasoline is required to include a minimum of 10% ethanol “Renewable Energy Standard.”
- Recent critical studies of lifecycle emissions by scientists (not funded by agri-business) indicate that ethanol produces 25% more greenhouse gases than gasoline – and has many other negative environmental impacts.
- 40% of US corn is grown to produce ethanol.



Ethanol vs Solar Panels

- Solar panels are 300 times more efficient than corn plants in capturing solar energy (photosynthesis 1-2% efficient) and converting to energy.
- Ethanol: high water and fertilizer use & competes with food raising prices
- Same land → 6,600 GW solar capacity
- **More electricity than 3× current US electricity generation!**
- Corn to cars is a dead end. Sun to grid is a climate solution.



Biden and Democrats on Climate



- Oil drilling: removed offshore areas but approved Willow Project and Alaska drilling
- Infrastructure Act & “Inflation Reduction Act” – largest and best climate legislation in US history – half way to 2050 Paris target:
 - Global cooperation
 - **Mitigation:** Specific targets for reducing greenhouse gas emissions including power plants and vehicles; promoted research, clean energy & electric vehicles
 - **Adaptation:** Enhance nation's ability to withstand the impacts of climate change through investments in infrastructure, community support, and disaster preparedness
 - **Climate Justice:** benefits for disadvantaged communities, community climate grants

2025-26 Going Backwards

In 2030, if current projections hold, the United States will drill for more oil and gas than at any point in its history. Russia and Saudi Arabia plan to do the same.

***Nations That Vowed to Halt Warming
Are Expanding Fossil Fuels, Report
Finds***

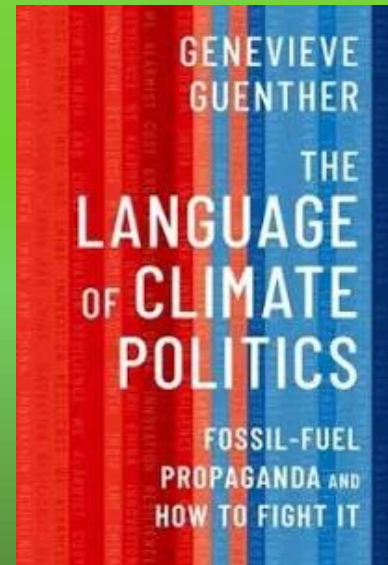
G20 poured more than \$1tn into fossil fuel subsidies despite Cop26 pledges – report

Trump and Republicans?

Ending all Biden efforts & everything possible to increase fossil fuel use and stop renewable energy!



- Withdrawing from the Paris Agreement – US only country not in
- Destroying climate science, data, and research, federal & university
- Dismantling the EPA and greenhouse gas regulation
- Discourage renewable energy, electric cars, and electrification
- Eliminating all previous Biden and blocking state climate actions
- Removing climate change from Defense Department planning
- Dismantling democracy and creating a climate of fear.



Climate Justice

Climate Crisis is a Justice Crisis

- Who caused the crisis? Who suffers first and most?
- Who knew and could have acted? Who fostered denial?
- How will decisions of the older generation impact the young?
- What is it like to be a climate victim or refugee? What responsibility do we have to victims and refugees?
- What should be America's role in the global community?
- What responsibilities do humans have to animals, plants, and the natural world?
- What would a just and fair approach to addressing the climate emergency look like? How do we create a just transition?

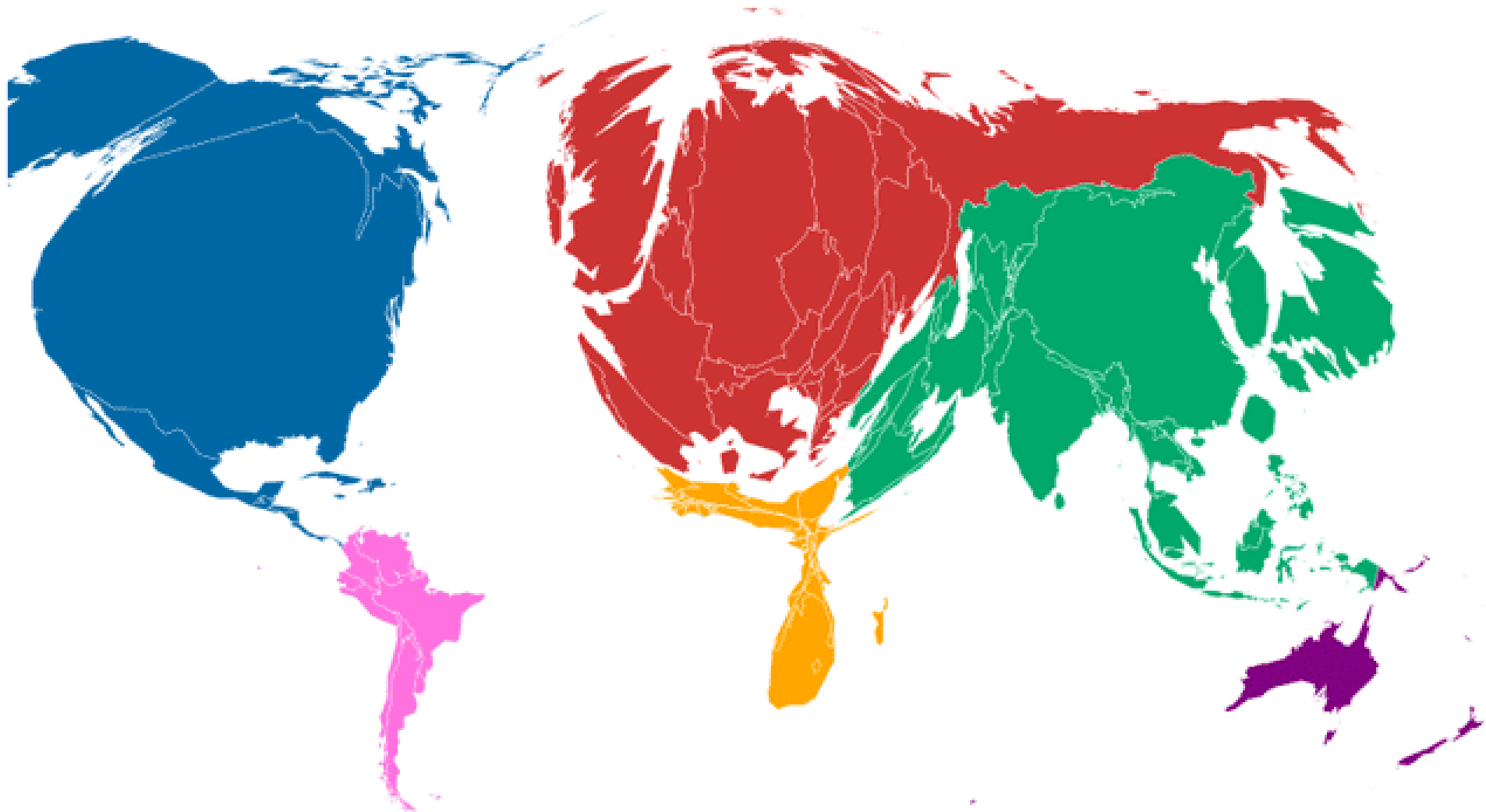


94% or 85% habitat loss
by 2050

Responsibility for Climate Crisis

BACKGROUND **RESPONSIBILITY** **VULNERABILITY**

🌐 Area Population Wealth Extraction Emissions Consumption **Historical** Reserves People at risk Sea level Poverty



About this map

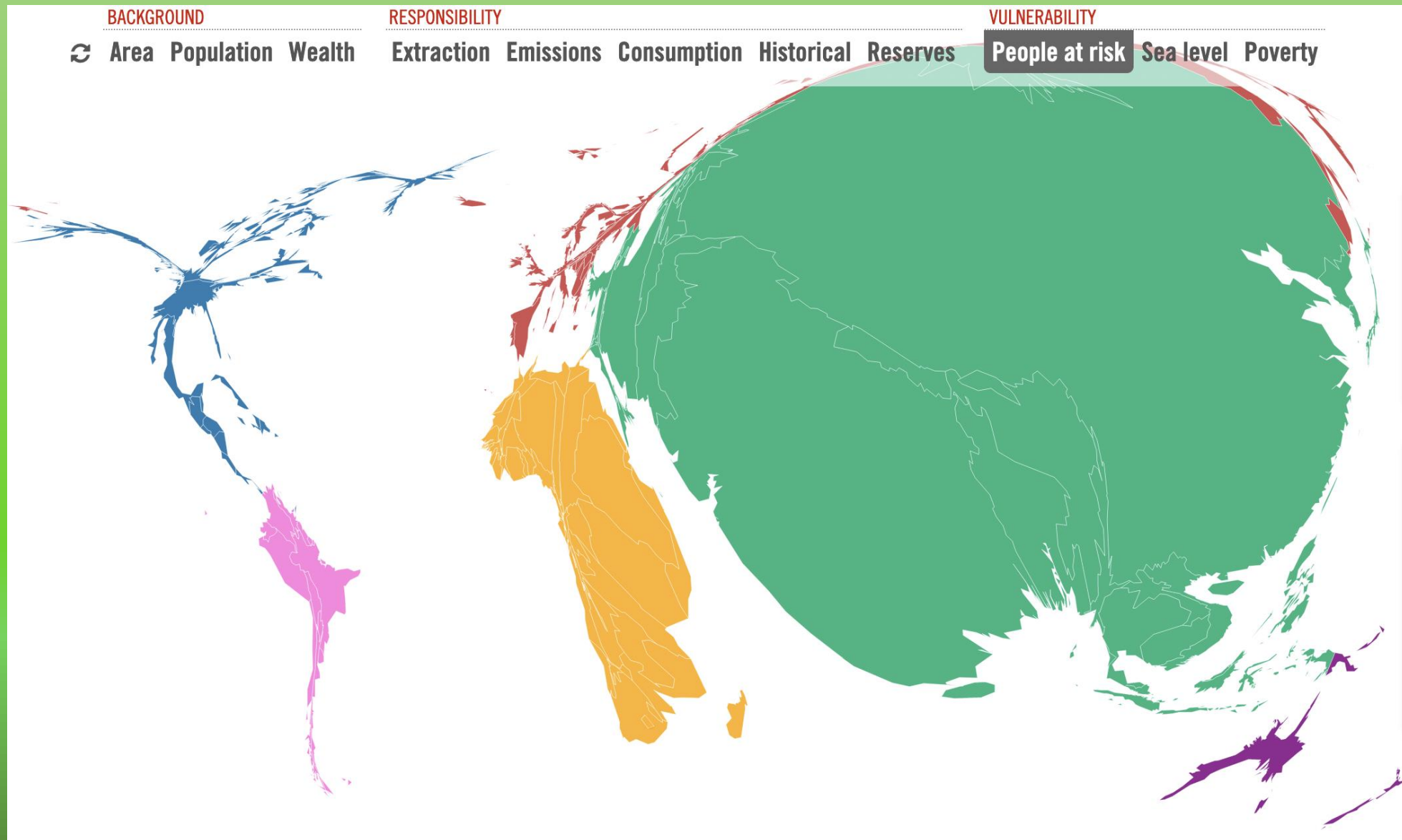
Country sizes show **CO2 emissions from energy use 1850–2007**. These historical (or 'cumulative') emissions remain relevant because CO2 can remain in the air for centuries. Europe and the US dominate, having released around half the CO2 ever emitted.

Shade this map by

Continents ▾

- Europe
- Africa
- Asia
- North America
- South America
- Oceania

Already Suffering from Global Heating



Justice of Climate Crisis Tied to History of Global Development & Underdevelopment

- Colonialism
- Slavery
- Capitalism
- Cold War
- Neoliberalism (radical privatization and deregulation creating state crisis and greater inequality)
- Racism, Segregation
- Rise of Carbon Economy



How Europe Underdeveloped Africa, Walter Rodney

Tropic of Chaos, Christian Parenti

This Changes Everything: Capitalism vs. the Climate, Naomi Klein

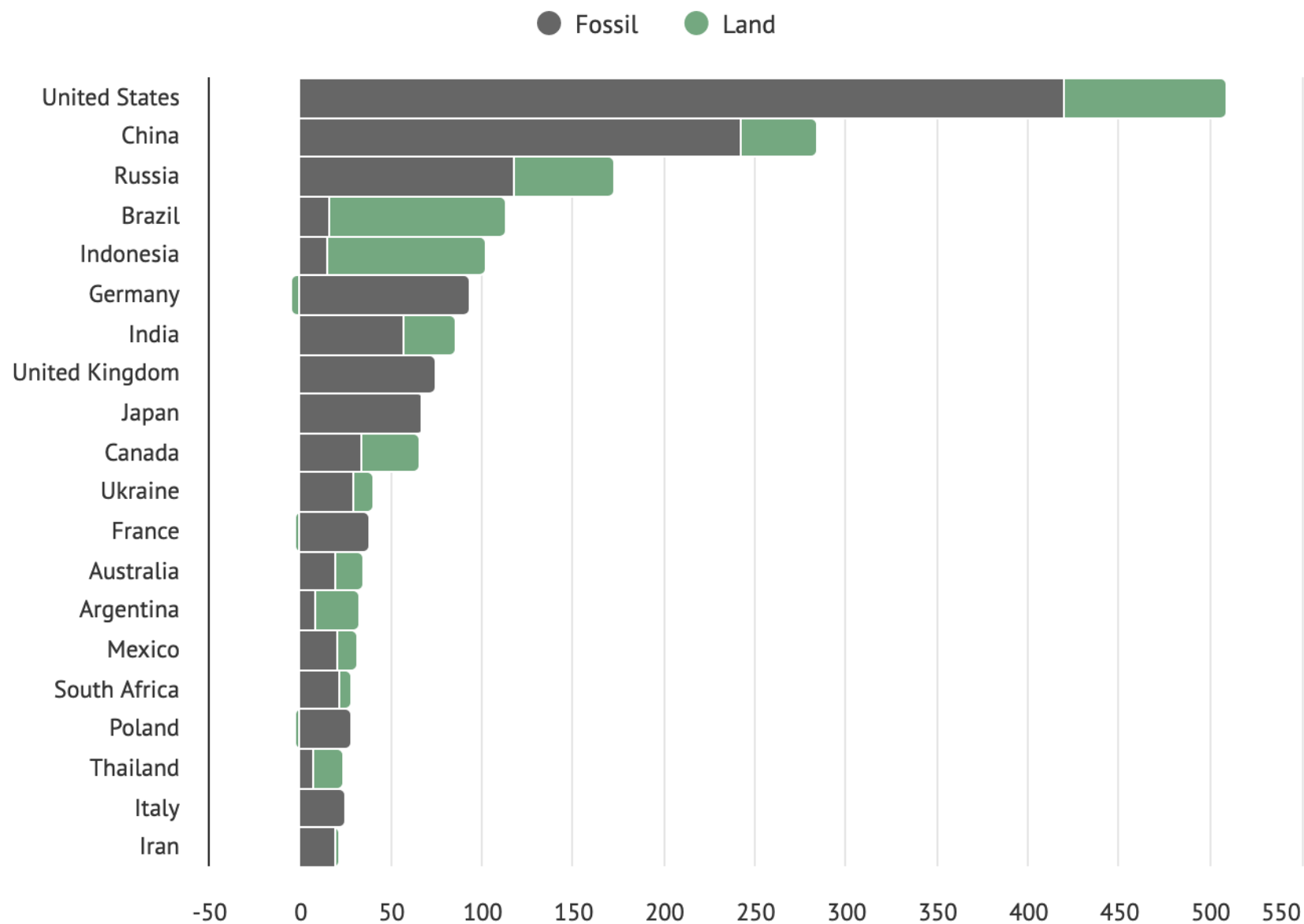
A Billion Black Anthropocenes or None, Kathryn Yusoff

How Capitalism Broke the Planet, Kate Arnoff

Nutmeg's Curse, Amitov Ghosh

The countries with the largest cumulative emissions 1850-2021

Billions of tonnes of CO2 from fossil fuels, cement, land use and forestry



Generational Justice



- While the present generation has a right to use the Earth and its natural resources to meet its own needs, the present generation must pass the Earth on to future generations in a condition no worse than they received it.
- “We do not inherit the world from our ancestors; we borrow it from our children.” -- common wisdom variously attributed to Chief Seattle, Ralph Waldo Emerson, Wendell Berry.
- Recognized in nearly all cultural and religious traditions, including the Judeo-Christian, Islamic, and Asian non-theistic traditions.
- A legal principle intergenerational equity is found in European common law, Islamic law, African, Native American, and many other indigenous laws and practices.
- Youth-led lawsuits: 1990s – Philippines , 2015 Pakistan & Netherlands, 2018 Columbia, 2019 Germany, France, Brazil, Argentina, and Turkey, 2021 Germany, Juliana v United States
- Generational justice should infuse and support all aspects of climate and environmental education and activism.

Climate Fascism

- Scapegoating immigrants and militarizing borders, leaders use fear and claim dictatorial power.
- Increasing inequality and poverty caused by climate change used to foster racism and a global politics of armed lifeboats.

But plants, animals, and humans have a right to migrate, and we must work together with all our brothers and sisters to protect the planet we share!

Carbon Footprints?

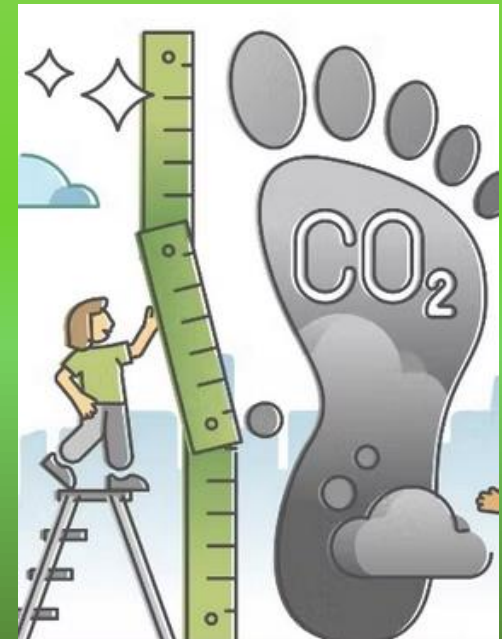
Why would fossil fuel companies promote the study of individual carbon footprints?

Toward a pedagogy of citizenship, not refined consumerism!



Understanding “Carbon Footprints”

- Footprints depend on energy source, not consumer behavior – electrify!
- **National** carbon footprints? (USA #1, 25-30% of total carbon)
- **Social class** carbon footprints / **Corporate** carbon footprints
- Environmental racism (inc. location of fossil fuel infrastructure)
- Challenges of carbon tax implementation
- Importance of laws, regulations, global cooperation
- Legal accountability, reparations, Green Fund



Fossil Fuel Companies Knew in the 1970s

- In the 1970s and 1980s Exxon, Mobil, Shell, Texaco and other oil company scientists were fully aware of what was going to happen with climate change – in fact their predictions for our time were spot on.
- In the late 1980s they starting spending hundreds of millions of dollars to lie, lobby the government, and confuse the public to deny and delay action. American Petroleum Institute (API)
- Bill McKibben calls this “The most consequential lie in human history.”
- In 2026 they are still lying and lobbying – and profiting.



American
Petroleum
Institute

Action Strategies

Local Analysis

Educate

Collaborate

Target Villains

Testify

Act Publicly

Local Analysis

- How is your location ground zero for climate change?
- How are those impacts experienced differently for different social, economic, and racial groups?
- What history made certain groups vulnerable? How does wealth or power cause or protect against the impacts? Who does the government help or protect? Why?
- What climate refugees are in your area and how are they treated?
- How are climate crises militarized in your area?
- What would need to be done to create a Just Transition in your community?



Meet with
community groups
and officials

Ask to speak at
meetings



Talk to other students



Talk with teachers
and administrators

Take Fear, Anxiety & Grief to Action

Recognizing that emotions are often what leads people to act, it is possible that feelings of ecological anxiety and grief, although uncomfortable, are in fact the crucible through which humanity must pass to harness the energy and conviction that are needed for the lifesaving changes now required.

(The Lancet, Cunsolo et al., 2020, p. 262)



Your Climate Persuasion Stories

- As a young person, your voice on climate change has special power.
- Climate stories are effective – there are also many other ways to communicate.
- You can have multiple stories. What do you want each climate story to convey?
- How might a collection of Ardea climate stories complement each other?
- How can you inspire others to find their climate story?

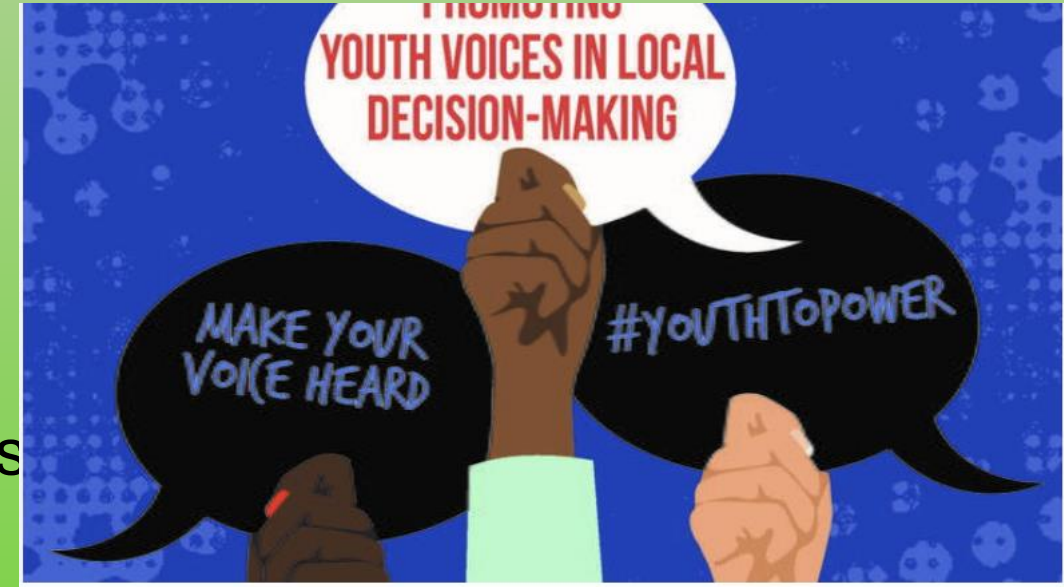
Suggestions for Climate Persuasion

- Emphasize your status as a young person
- Express concern and fear about the future
- Recognize the urgency of action
- Find a way to connect your thinking to issues that will touch your audience – awareness of nature is fine, but consider societal impacts
- Show your knowledge about climate change and specific strategies to address it
- Modify your story and focus for specific audiences



Community Audiences

- General public
 - Teachers, school boards, service districts youth programs
 - Townships, city, local government - Do they have a climate action plan? What are they doing to reduce greenhouse gas emissions?
 - Homeowners - access to insulation & solar panels / Realtors – have a carbon rating for homes
 - Utility boards / Businesses – car dealers, local better business bureau, sustainable business forums
- & Don't forget online audiences: other young people, potential voters, etc





Time for maximum
press coverage

Go to the street



Climate activists host 'die in' on Black Friday at Michigan mall

Updated: Nov. 29, 2019, 3:46 p.m. | Published: Nov. 29, 2019, 3:45 p.m.



Michigan ▾



Demonstrators pretend to be dead during Black Friday at The Crossroads mall in Portage.

Advertisement





How banks have fueled climate change



JPMorgan Chase Leads the Way (to Climate Chaos)

In the three years since the Paris Agreement, JPMorgan Chase was the:

#1

Banker of Fossil Fuels (BY 29%)

\$196 B TOTAL

#1 in 2016, 2017, and 2018

#1

Banker of 100 Top Companies Expanding Fossil Fuels (BY 68%)

#1 in 2016, 2017, and 2018



#1 U.S. Banker of Tar Sands Oil



#1 Banker of Arctic Oil & Gas



#1 Banker of Ultra-Deepwater Oil & Gas



#2 Banker of Fracking (JUST BEHIND WELLS FARGO)



#1 Banker of LNG



#1 U.S. Banker of Coal Mining

Cut Emissions
Not Trees



I sequester carbon

I turn CO_2 to Oxygen

I ♥
Solar Energy

I'm rooting
for the
EARTH



I'm
Growing to
help

I can't run from
climate change.
Can you?

Informational Hub



Create your own shirts
Social media photo booth



XR “Beach Party”

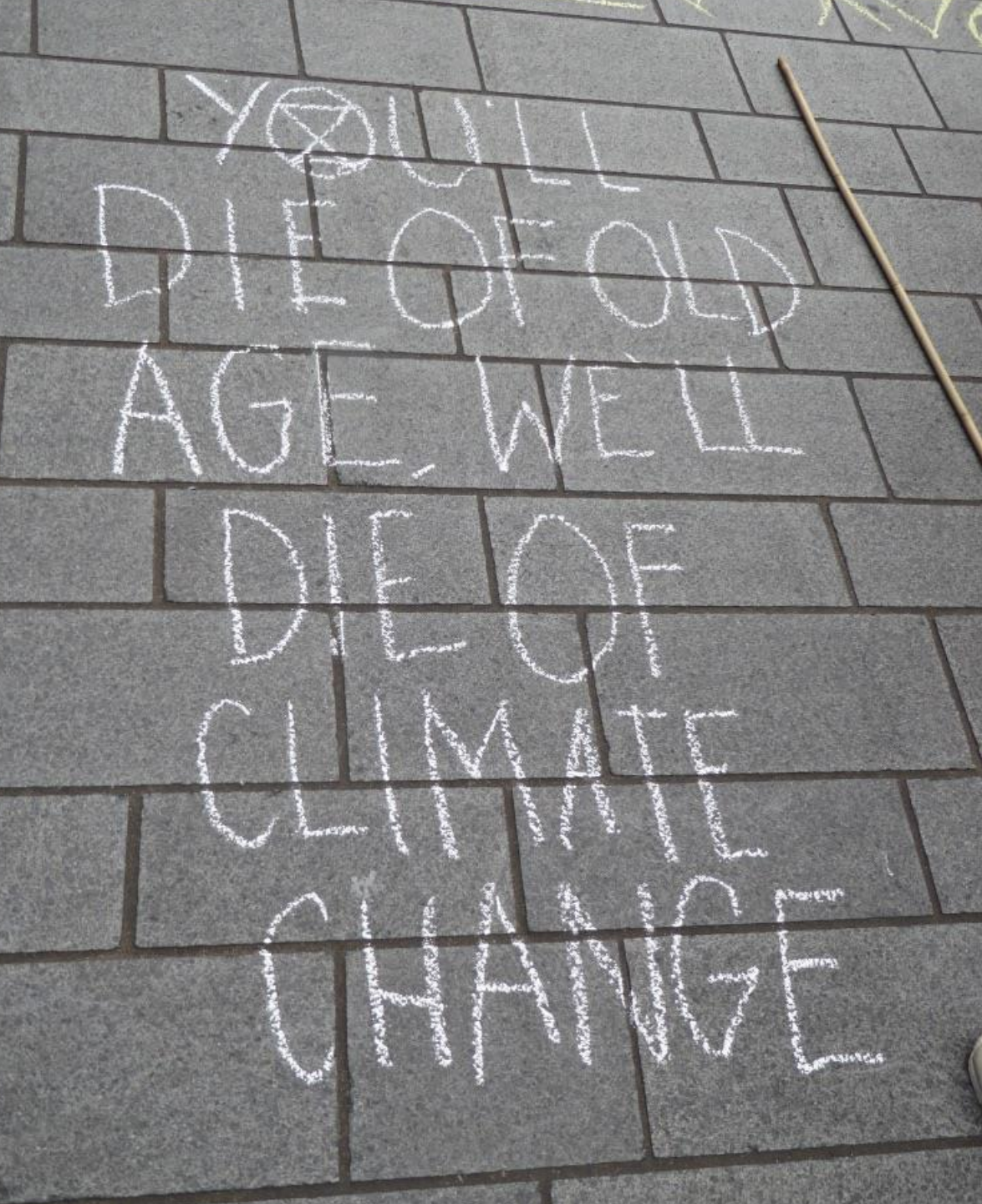


Painted across the city



Drown In







BBC SpooF Performance

BBC Blockage





Public Protest

- rallies and marches
- mock funerals for the earth
- fake blood / chalking to represent environmental threats to children's future



Civil Disobedience

- climbing buildings or bridges to unfurl banners
- blocking streets and bridges in front of government offices, banks, or industries causing climate change
- chaining or gluing themselves down

Presentation
online at:



<https://allen-webb-wmu.github.io/website/ClimateAdea2025.pdf>

Extra Slides

Carbon Taxes

- A **high, durable carbon price** is still the economist's ideal—but is **rare** in practice. Australia repealed its carbon price in 2014; France's proposed fuel tax hikes triggered the “gilets jaunes” backlash; and in **2025 Canada set its federal consumer carbon price rates to zero**, despite years of rebates. It is hard to **pass, raise, and keep** broad fuel taxes.
- **Renewables + grids + standards + incentives** are doing most of the near-term heavy lifting, because they're **politically passable** and **cost-competitive today**.
- The **best-performing countries/regions** blend tools: **price carbon where you can, invest and regulate where you must**—and always build the wires and flexibility to use the clean power you're buying.

Carbon Capture and Storage

Investing in carbon capture and direct air capture increases air pollution, carbon dioxide emissions, fossil-fuel mining, fossil-fuel infrastructure, carbon dioxide and fossil-fuel pipelines, energy requirements, private energy costs, and social energy costs ([by a factor of 9-12](#)) relative to investing the same money in clean, renewable energy.

What is more, over 82 percent of all carbon dioxide captured worldwide, in reality, is used for enhanced oil recovery. During this process, 30 to 40 percent of the carbon dioxide captured is released back to the air.

Further, another 20 to 80 percent captured is released due to the additional oil obtained, so a total of [50 to 120 percent](#) of the carbon dioxide captured during enhanced oil recovery is released.

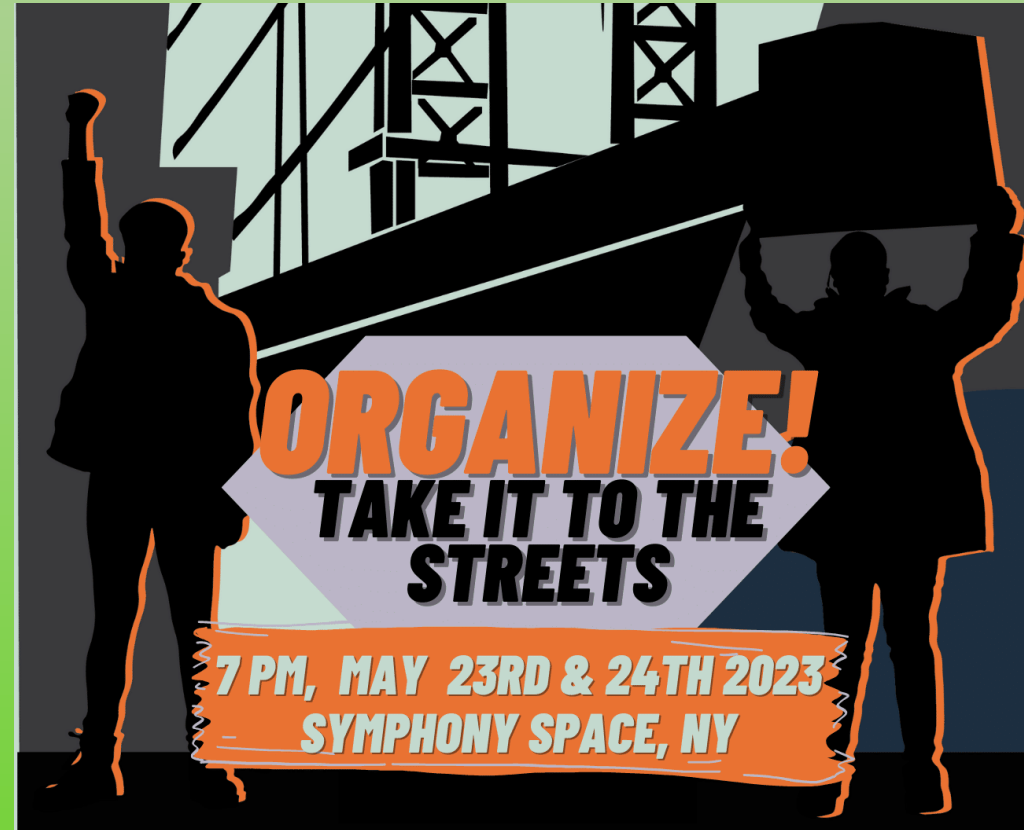
Hydrogen and “Green” Hydrogen

- Using renewable energy to make hydrogen is a lot less efficient than simply using the renewable energy directly.
- Releasing hydrogen... increases the lifetime of methane in the atmosphere with huge climate impact & hydrogen is the leakiest gas known to humankind.

Stages of Activism

3.5% active and sustained participation of population

- Planning
- Trigger Event
- Disruption, Sacrifice, Escalation
- Whirlwind
- Institutionalize



Overton Window

- Unthinkable ideas can actually become the basis for government policy.
- Social movements and uprisings may make their greatest impacts as they change broad public understanding of what is happening and acceptable to do about it.
- Overton suggested that ideas can move in a sequence from unthinkable to radical to acceptable to sensible to popular to policy.

